Research development summary

Trace Sparsity and Arithmeticity for Compact Hyperbolic Surfaces

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1 Scope and available records

This is an abridged retrospective account of the development recorded in the source project's provenance ledger, audit log, and mathematical notes. It accompanies the focused compact trace-sparsity manuscript. The summary reports that recorded account; it does not independently reconstruct a chronology from original messages.

The principal records are `PROVENANCE_INFLUENCE_LEDGER.md`, `AUDIT_LOG.md`, `CRITICAL_PATH_AUDIT.md`, and `FINAL_ARITHMETICITY_DEDUCTION.md`. Supporting explanations appear in `SPECTRAL_COUNTING_BRIDGE.md` and `LOCAL_SQUARE_DENSITY_PROOF.md`. The accompanying provenance document identifies their source revision.

No original conversation export was available in the audited project tree. This summary therefore contains no verbatim model quotations, reconstructed dialogue, or authenticated original prompt. The available release records do not establish a model version, runtime, token count, or compute budget. The development described here belongs to an external model-assisted project.

2 Problem brief

The manuscript studies the signed trace set of a cocompact Fuchsian lattice and its counting function

$$N_\Gamma(X) = \#(T(\Gamma) \cap [-X, X]).$$

Its main claim is arithmeticity under the critical bound

$$\limsup_{X\to\infty} \frac{\log N_\Gamma(X)}{\log X} \leq 1.$$

Literal linear growth implies this bound, while bounded clustering implies linear growth by covering a long interval with bounded-length intervals. The compact Schmutz and Sarnak conclusions are consequences of the stated critical-growth theorem. This is a description of the submitted manuscript's claim, not a report of independent mathematical certification.

3 Development described by the project records

Initial framework and approaches that did not close the argument

The provenance ledger identifies signed traces, the subgroup generated by squares, trace fields, quaternion algebras, and the trace-of-powers recurrence as inherited structural ingredients. It describes projective square-root calculations as an early source of two-trace identities.

The records then distinguish several limitations. Difference-set density can produce arbitrarily close pairs without contradicting bounded clustering. Scalar gradings and additive-energy arguments do not by themselves recover the needed arithmetic structure. A route through a universal radical lower bound for one fixed recurrence would entail strong unresolved statements of Mersenne and Lucas–Wieferich type. These are recorded obstructions to particular approaches, not additional assumptions of the focused theorem.

The moving-element change of strategy

The ledger identifies fixing a Chebyshev recurrence index N while varying the group element as the decisive change. The manuscript uses elements of the form $a_m = b^m h_m$, with a short perturbation h_m. The aim is to retain the relevant local expansion while keeping the prime-power depths of the recurrence ideals bounded.

The documented argument combines four stages:

1. A reduced-divisor application of Roth's theorem supplies a lower bound with coefficient $2N-4-\epsilon$.
2. A local prime-square density estimate and bounded-power expansion are used in the selection of the varying elements.
3. A uniform two-trace fiber bound controls the multiplicity with which group elements represent a trace-graph edge.
4. Pruning the trace graph, congruence counting, and an injection for length-two paths supply the complementary upper estimate.

The notes emphasize the parameter order: first let m tend to infinity, then let the auxiliary errors tend to zero, and finally let N tend to infinity. The claimed resulting estimate compares the local expansion functional at an algebraic specialization y with length at the original character x:

$$\mathcal{J}_{k_y}(\operatorname{tr}_y b) \leq \frac{\ell_x(b)}{2}.$$

The manuscript contains the mathematical statements, uniformity ranges, and proofs used in these stages.

Arithmeticity and specialization

At an algebraic character, the distinguished real place already contributes half the translation length. The manuscript argues that the upper estimate forces all other nonnegative local contributions to vanish. Its final deduction applies Takeuchi's criterion to the finite-index subgroup generated by squares and then uses commensurability.

For a transcendental character, the records describe algebraic Fuchsian specializations preserving the relevant trace collisions. The claimed local estimate gives marked-length domination; Thurston's separating asymmetric metric is then used to force equality of the characters. The passage to compact orbifolds uses a torsion-free finite-index subgroup.

4 Checks and repairs recorded in the project

The audit log records repeated mathematical, reference, exposition, and formal-scope checks. This summary reports those entries without establishing that they constitute independent human peer review. Examples of the recorded repairs include:

- retaining degree factors when the field used in the local expansion functional changes;
- separating the zero-norm branch of the two-trace calculation from the quadratic divisor argument;
- displaying the modulus-versus-radius condition needed to absorb the congruence-counting error and choosing constants in the stated order;
- including finite covolume and finite index in the Takeuchi deduction; and
- separating literal linear growth from the weaker critical-growth hypothesis and checking the exact order of the limits.

5 What the accompanying Lean project establishes

The Lean package proves selected recurrence and polynomial identities, finite combinatorial inequalities, and limiting coefficient statements. It also checks conditional deductions from explicitly supplied analytic, geometric, specialization, and arithmetic hypotheses.

The package does not construct the relevant hyperbolic surfaces or formalize all of the manuscript's external inputs. Its public arithmeticity implications use abstract hypotheses. Finite consistency examples and counterexamples audit those hypotheses and their numerical translation; they are not geometric realizations. A successful build or an axiom report containing only standard logical principles does not by itself establish that all the geometric hypotheses have been supplied. The manuscript map gives the declaration-level boundary.

6 Account metadata and publication boundary

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Raw conversation exports are not included. Auditing any later transcript excerpts would require reviewing those actual records and their hidden metadata. The document above is a summary of available project records, not an assertion that an unavailable conversation archive has been anonymized or authenticated.